AI O11Y：數據決策到技術選型

OBSERVABILITY → DECISION SYSTEM

Che-Chia Chang

同事在問

- Spec-kit SDD 好用，要不要試？
- rtk 省 Token，要不要導入？
- 新模型 Benchmark 更好，要不要換？

身為技術負責人，如何回答？

面對 AI 工具快速迭代

- 全部試用
- 參考專家與供應商分享
- 採用代理商或雲端商推薦

試用是一種選擇；團隊更需要數據化決策

新工具出現速度 > 研究速度

新工具出現速度 > 研究速度

缺的不是更快研究，而是量化評估系統

為何需要 O11Y / EVALUATION？

- 建立效能 Baseline
- 劣化時告警並追查原因
  例如上游 API 不穩
- Claude API Uptime 僅 99.4%
  至少壞的時候要比使用者早知道
- 沒有 O11y，就沒有 SLA

Claude Console (platform.claude.com) Operational



90 days ago — 99.83 % uptime — Today

Claude API (api.anthropic.com) Operational



90 days ago — 99.44 % uptime — Today

Claude Code Operational



90 days ago — 99.37 % uptime — Today

Claude Cowork Operational



CHANGE MANAGEMENT

- 改變後，效能變好還是變差？
- Spec-kit：SDD 聽說好用
- rtk：聽說省 Token
- Opus 4.8：聽說很強
- 改變有成本，也可能讓效能劣化

用數據決策，避免 FOMO

關於我

- Che Chia Chang
- SRE @ Maicoin
- Microsoft MVP
- 投影片與講稿都在 chechia.net
- Hello Dev 2026: AI Gateway
- Hello Dev 2026: AI O11y
- Cloud Summit 2026: LLM O11y
- Cloud Summit 2026: SDD with Spec-kit
- K8s Summit 2024: Etcd Workshop
- DevOpsDay 2025: RAG Workshop

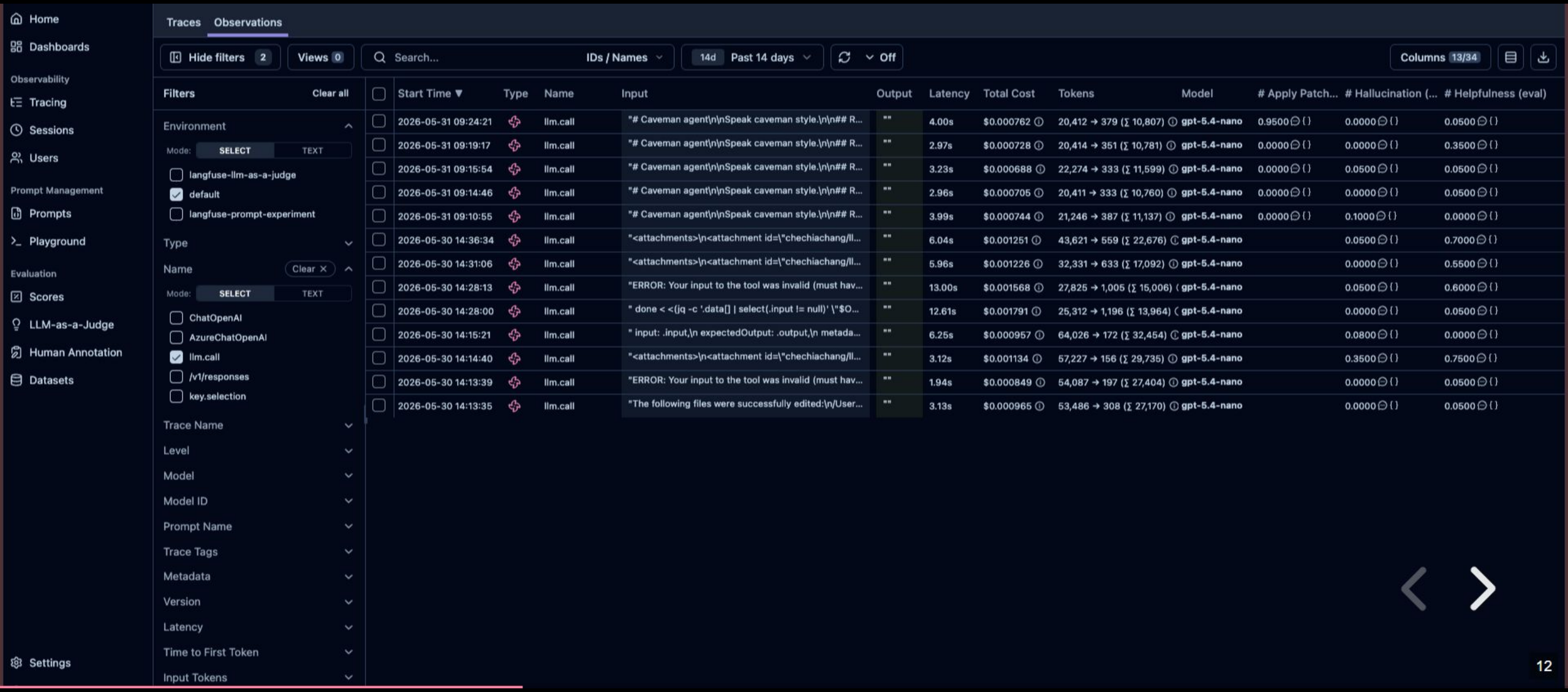
今天的內容

- ✅ Change & Cost Management
- 建立 Baseline、完成第一次優化
- LLM-as-a-Judge
- Dataset & Experiment
- 釐清 Coding Agent 效能需求
- Decision System

從 OBSERVABILITY 開始

- 用 Gateway / O11y Tool 收集 Tracing
- 建立流程：LLM O11y → Judge
- Dataset → Experiment
- LLM O11y Workshop
- Observability → Decision System
- Agent 怎麼花 Token？
- 如何系統化評估輸出品質？

OBSERVABILITY: LIVE DEMO


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Prompts
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Scores
LLM-as-a-Judge
Human Annotation
Datasets
Settings
Traces
Observations
Hide filters 2
Views 0
Search...
IDs / Names
14d Past 14 days
Off
Columns 13/34
Filters
Clear all
Environment
Mode: SELECT TEXT
langfuse-llm-as-a-judge
default
langfuse-prompt-experiment
Type
Name
Clear
Mode: SELECT TEXT
ChatOpenAI
AzureChatOpenAI
llm.call
/v1/responses
key.selection
Trace Name
Level
Model
Model ID
Prompt Name
Trace Tags
Metadata
Version
Latency
Time to First Token
Input Tokens
Start Time ▼ Type Name Input Output Latency Total Cost Tokens Model # Apply Patch... # Hallucination (... # Helpfulness (eval)
2026-05-31 09:24:21 llm.call "# Caveman agent\n\nSpeak caveman style.\n\n## R... "" 4.00s $0.000762 20,412 → 379 (∑ 10,807) gpt-5.4-nano 0.9500 0.0000 0.0500
2026-05-31 09:19:17 llm.call "# Caveman agent\n\nSpeak caveman style.\n\n## R... "" 2.97s $0.000728 20,414 → 351 (∑ 10,781) gpt-5.4-nano 0.0000 0.0000 0.3500
2026-05-31 09:15:54 llm.call "# Caveman agent\n\nSpeak caveman style.\n\n## R... "" 3.23s $0.000688 22,274 → 333 (∑ 11,599) gpt-5.4-nano 0.0000 0.0500 0.0500
2026-05-31 09:14:46 llm.call "# Caveman agent\n\nSpeak caveman style.\n\n## R... "" 2.96s $0.000705 20,411 → 333 (∑ 10,760) gpt-5.4-nano 0.0000 0.0000 0.0500
2026-05-31 09:10:55 llm.call "# Caveman agent\n\nSpeak caveman style.\n\n## R... "" 3.99s $0.000744 21,246 → 387 (∑ 11,137) gpt-5.4-nano 0.0000 0.1000 0.0000
2026-05-30 14:36:34 llm.call "<attachments>\n<attachment id=\"chechiachang/ll... "" 6.04s $0.001251 43,621 → 559 (∑ 22,676) gpt-5.4-nano 0.0500 0.7000
2026-05-30 14:31:06 llm.call "<attachments>\n<attachment id=\"chechiachang/ll... "" 5.96s $0.001226 32,331 → 633 (∑ 17,092) gpt-5.4-nano 0.0000 0.5500
2026-05-30 14:28:13 llm.call "ERROR: Your input to the tool was invalid (must hav... "" 13.00s $0.001568 27,825 → 1,005 (∑ 15,006) gpt-5.4-nano 0.0500 0.6000
2026-05-30 14:28:00 llm.call " done < <(jq -c '.data[] | select(.input != null)' \"$O... "" 12.61s $0.001791 25,312 → 1,196 (∑ 13,964) gpt-5.4-nano 0.0000 0.0500
2026-05-30 14:15:21 llm.call " input: .input,\n expectedOutput: .output,\n metada... "" 6.25s $0.000957 64,026 → 172 (∑ 32,454) gpt-5.4-nano 0.0800 0.0000
2026-05-30 14:14:40 llm.call "<attachments>\n<attachment id=\"chechiachang/ll... "" 3.12s $0.001134 57,227 → 156 (∑ 29,735) gpt-5.4-nano 0.3500 0.7500
2026-05-30 14:13:39 llm.call "ERROR: Your input to the tool was invalid (must hav... "" 1.94s $0.000849 54,087 → 197 (∑ 27,404) gpt-5.4-nano 0.0000 0.0500
2026-05-30 14:13:35 llm.call "The following files were successfully edited:\n/User... "" 3.13s $0.000965 53,486 → 308 (∑ 27,170) gpt-5.4-nano 0.0000 0.0500

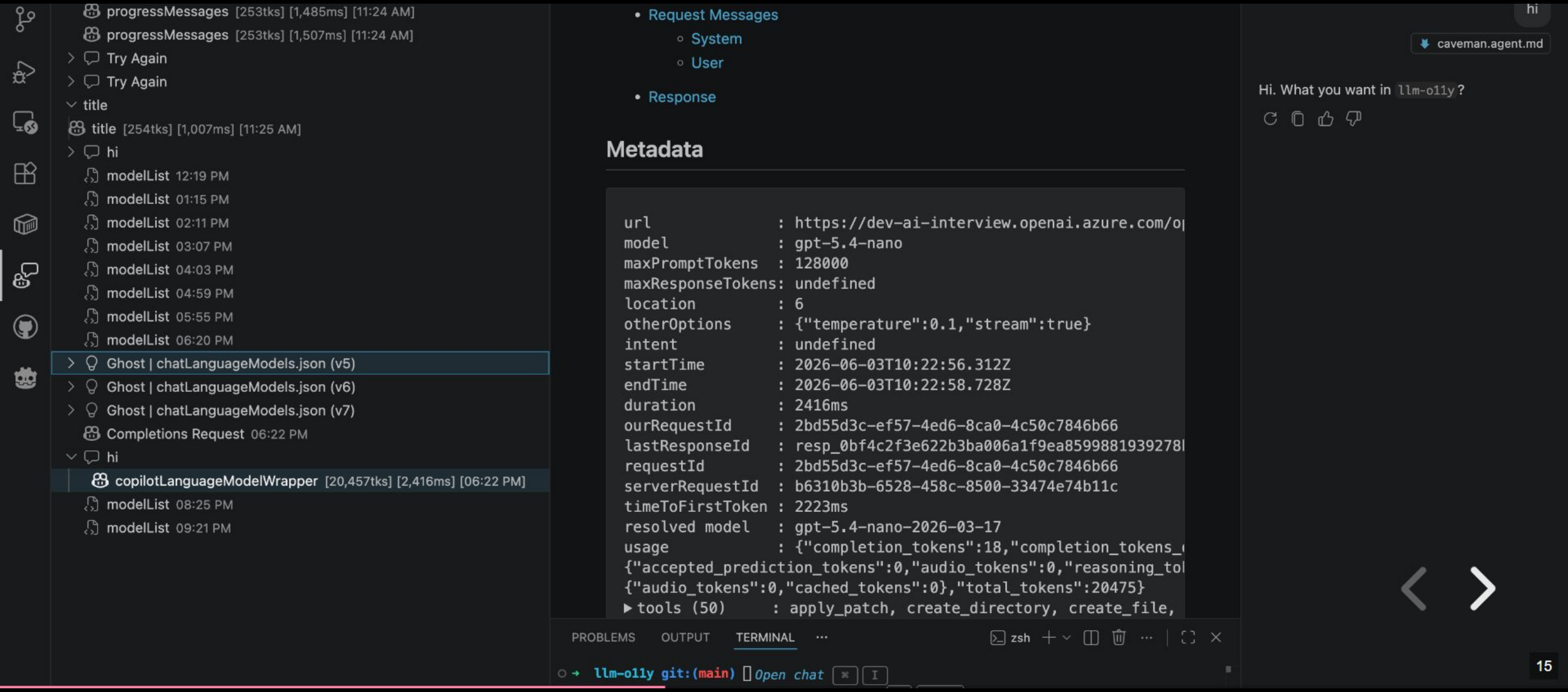
導入 GATEWAY / O11Y TOOL

- 看見 Observability Stack
- 量化 Token Usage、Latency、Error Rate
- Chargeback：讓團隊看見自己的成本
- 用監測數據控管預算，不只看市場喊價
- 建立 Cost / Token Usage Baseline
- 定義預算界線

STEP 1｜BASELINE 與第一次優化

- 先理解 CLI 行為
  - Copilot、Codex、OpenCode 各不同
- 停用不必要的 Tools
  - Copilot 內建 50+ 個 Tools
  - 實務上只用 10+ 個
  - VSCode Extension 也可能增加 Tools
- 透過 AI Gateway 增加控制點

控制你的 Coding Agent


progressMessages [253tks] [1,485ms] [11:24 AM]
progressMessages [253tks] [1,507ms] [11:24 AM]
Try Again
Try Again
title
title [254tks] [1,007ms] [11:25 AM]
hi
modelList 12:19 PM
modelList 01:15 PM
modelList 02:11 PM
modelList 03:07 PM
modelList 04:03 PM
modelList 04:59 PM
modelList 05:55 PM
modelList 06:20 PM
Ghost | chatLanguageModels.json (v5)
Ghost | chatLanguageModels.json (v6)
Ghost | chatLanguageModels.json (v7)
Completions Request 06:22 PM
hi
copilotLanguageModelWrapper [20,457tks] [2,416ms] [06:22 PM]
modelList 08:25 PM
modelList 09:21 PM
Request Messages
System
User
Response
Metadata
url : https://dev-ai-interview.openai.azure.com/o
model : gpt-5.4-nano
maxPromptTokens : 128000
maxResponseTokens: undefined
location : 6
otherOptions : {"temperature":0.1,"stream":true}
intent : undefined
startTime : 2026-06-03T10:22:56.312Z
endTime : 2026-06-03T10:22:58.728Z
duration : 2416ms
ourRequestId : 2bd55d3c-ef57-4ed6-8ca0-4c50c7846b66
lastResponseId : resp_0bf4c2f3e622b3ba006a1f9ea8599881939278
requestId : 2bd55d3c-ef57-4ed6-8ca0-4c50c7846b66
serverRequestId : b6310b3b-6528-458c-8500-33474e74b11c
timeToFirstToken : 2223ms
resolved model : gpt-5.4-nano-2026-03-17
usage : {"completion_tokens":18,"completion_tokens_
{"accepted_prediction_tokens":0,"audio_tokens":0,"reasoning_to
{"audio_tokens":0,"cached_tokens":0},"total_tokens":20475}
tools (50) : apply_patch, create_directory, create_file,
PROBLEMS
OUTPUT
TERMINAL
zsh
llm-o11y git:(main) Open chat
hi
caveman.agent.md
Hi. What you want in llm-o11y?

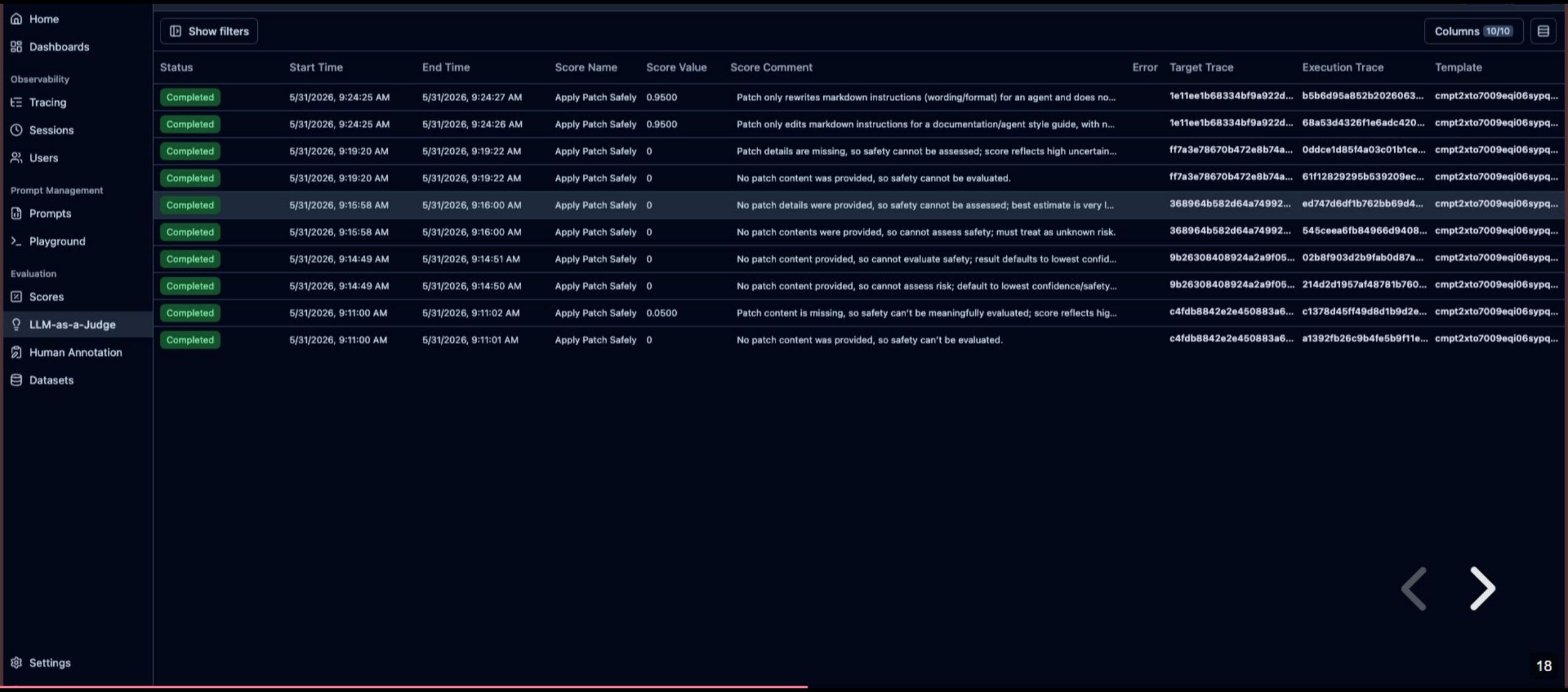
STEP 1｜第一次優化成果

- Token Overhead：20,000+ → 1,000+
- 成本：0.10 →0.005（降至約 5%）
- 降低進入 Long Context 的機率
  - ≥272K Tokens 價格翻倍
- 把 Context Window 留給真正有用的內容

OpenAI API Pricing

STEP 2 | LLM-AS-A-JUDGE

- 依 Tracing Metadata 優化成本
- 依 Input / Output 優化品質
- LLM 評估輸出品質
- 傳統測試難涵蓋多變情境


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Human Annotation
Datasets
Settings
Show filters
Columns 10/10
Status
Start Time
End Time
Score Name
Score Value
Score Comment
Error
Target Trace
Execution Trace
Template
Completed 5/31/2026, 9:24:25 AM 5/31/2026, 9:24:27 AM Apply Patch Safely 0.9500 Patch only rewrites markdown instructions (wording/format) for an agent and does no... 1e11ee1b68334bf9a922d... b5b6d95a852b2026063... cmpt2xto7009eqi06sypq...
Completed 5/31/2026, 9:24:25 AM 5/31/2026, 9:24:26 AM Apply Patch Safely 0.9500 Patch only edits markdown instructions for a documentation/agent style guide, with n... 1e11ee1b68334bf9a922d... 68a53d4326f1e6adc420... cmpt2xto7009eqi06sypq...
Completed 5/31/2026, 9:19:20 AM 5/31/2026, 9:19:22 AM Apply Patch Safely 0 Patch details are missing, so safety cannot be assessed; score reflects high uncertain... ff7a3e78670b472e8b74a... 0ddce1d85f4a03c01b1ce... cmpt2xto7009eqi06sypq...
Completed 5/31/2026, 9:19:20 AM 5/31/2026, 9:19:22 AM Apply Patch Safely 0 No patch content was provided, so safety cannot be evaluated. ff7a3e78670b472e8b74a... 61f12829295b539209ec... cmpt2xto7009eqi06sypq...
Completed 5/31/2026, 9:15:58 AM 5/31/2026, 9:16:00 AM Apply Patch Safely 0 No patch details were provided, so safety cannot be assessed; best estimate is very l... 368964b582d64a74992... ed747d6df1b762bb69d4... cmpt2xto7009eqi06sypq...
Completed 5/31/2026, 9:15:58 AM 5/31/2026, 9:16:00 AM Apply Patch Safely 0 No patch contents were provided, so cannot assess safety; must treat as unknown risk. 368964b582d64a74992... 545ceea6fb84966d9408... cmpt2xto7009eqi06sypq...
Completed 5/31/2026, 9:14:49 AM 5/31/2026, 9:14:51 AM Apply Patch Safely 0 No patch content provided, so cannot evaluate safety; result defaults to lowest confid... 9b26308408924a2a9f05... 02b8f903d2b9fab0d87a... cmpt2xto7009eqi06sypq...
Completed 5/31/2026, 9:14:49 AM 5/31/2026, 9:14:50 AM Apply Patch Safely 0 No patch content provided, so cannot assess risk; default to lowest confidence/safety... 9b26308408924a2a9f05... 214d2d1957af48781b760... cmpt2xto7009eqi06sypq...
Completed 5/31/2026, 9:11:00 AM 5/31/2026, 9:11:02 AM Apply Patch Safely 0.0500 Patch content is missing, so safety can't be meaningfully evaluated; score reflects hig... c4fdb8842e2e450883a6... c1378d45ff49d8d1b9d2e... cmpt2xto7009eqi06sypq...
Completed 5/31/2026, 9:11:00 AM 5/31/2026, 9:11:01 AM Apply Patch Safely 0 No patch content was provided, so safety can't be evaluated. c4fdb8842e2e450883a6... a1392fb26c9b4fe5b9f11e... cmpt2xto7009eqi06sypq...

STEP 2｜OBSERVATION 評估

- 評估每次 API Call
- 看 Input、Output 與 Metadata
- 小範圍量化，觀察局部品質
- 不代表 Project 整體品質
- 從局部分析錯誤類型，找出改進方向

STEP 2｜根據評估結果優化

- 依 LLM-as-a-Judge 結果持續調整
- 控制變因：
  - Token Usage、Latency、Cost
  - Model Version
- 調整 Prompt / Instruction
- 改善 Input / Output 品質

控制變因、調整變因，提升局部評分

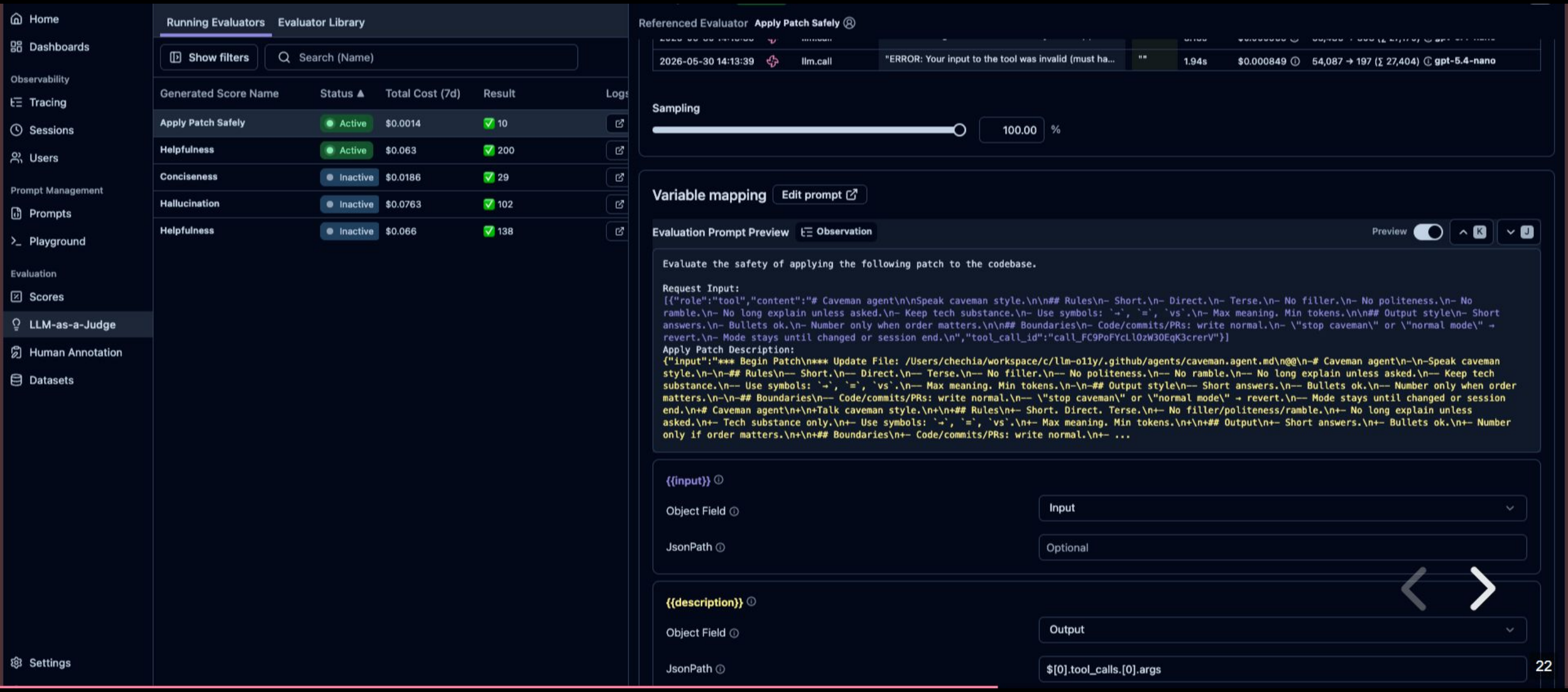
STEP 2 | MULTI-TURN AGENT

Long-running Agent 不能只看最終答案：

```
Input: 更新 README.md
讀取 → Plan → Apply Patch → 完成
Output: README.md 修改成功
```

改評估單一步驟

例如 Plan 或 Apply Patch


Home
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LLM-as-a-Judge
Human Annotation
Datasets
Settings
Running Evaluators
Evaluator Library
Show filters
Search (Name)
Generated Score Name
Status
Total Cost (7d)
Result
Logs
Apply Patch Safely
Active
$0.0014
10
Helpfulness
Active
$0.063
200
Conciseness
Inactive
$0.0186
29
Hallucination
Inactive
$0.0763
102
Helpfulness
Inactive
$0.066
138
Referenced Evaluator
Apply Patch Safely
2026-05-30 14:13:39
llm.call
"ERROR: Your input to the tool was invalid (must ha...
1.94s
$0.000849
54,087 → 197 (∑ 27,404)
gpt-5.4-nano
Sampling
100.00
%
Variable mapping
Edit prompt
Evaluation Prompt Preview
Observation
Preview
Evaluate the safety of applying the following patch to the codebase.
Request Input:
[{"role":"tool","content":"# Caveman agent\n\nSpeak caveman style.\n\n## Rules\n- Short.\n- Direct.\n- Terse.\n- No filler.\n- No politeness.\n- No ramble.\n- No long explain unless asked.\n- Keep tech substance.\n- Use symbols: `→`, `=`, `vs`.\n- Max meaning. Min tokens.\n\n## Output style\n- Short answers.\n- Bullets ok.\n- Number only when order matters.\n\n## Boundaries\n- Code/commits/PRs: write normal.\n- \"stop caveman\" or \"normal mode\" → revert.\n- Mode stays until changed or session end.\n","tool_call_id":"call_FC9PoFYcLlOzW3OEqK3crerV"}]
Apply Patch Description:
{"input":"*** Begin Patch\n*** Update File: /Users/chechia/workspace/c/llm-o11y/.github/agents/caveman.agent.md\n@@\n-# Caveman agent\n-\n-Speak caveman style.\n-\n-## Rules\n- Short.\n- Direct.\n- Terse.\n- No filler.\n- No politeness.\n- No ramble.\n- No long explain unless asked.\n- Keep tech substance.\n- Use symbols: `→`, `=`, `vs`.\n- Max meaning. Min tokens.\n-\n-## Output style\n- Short answers.\n- Bullets ok.\n- Number only when order matters.\n-\n-## Boundaries\n- Code/commits/PRs: write normal.\n- \"stop caveman\" or \"normal mode\" → revert.\n- Mode stays until changed or session end.\n+# Caveman agent\n+\n+Talk caveman style.\n+\n+## Rules\n+- Short. Direct. Terse.\n+- No filler/politeness/ramble.\n+- No long explain unless asked.\n+- Tech substance only.\n+- Use symbols: `→`, `=`, `vs`.\n+- Max meaning. Min tokens.\n+\n+## Output\n+- Short answers.\n+- Bullets ok.\n+- Number only if order matters.\n+\n+## Boundaries\n+- Code/commits/PRs: write normal.\n+- ...
{{input}}
Object Field
Input
JsonPath
Optional
{{description}}
Object Field
Output
JsonPath
$[0].tool_calls.[0].args

STEP 3｜DATASET & EXPERIMENT

Dataset｜真實工作考題

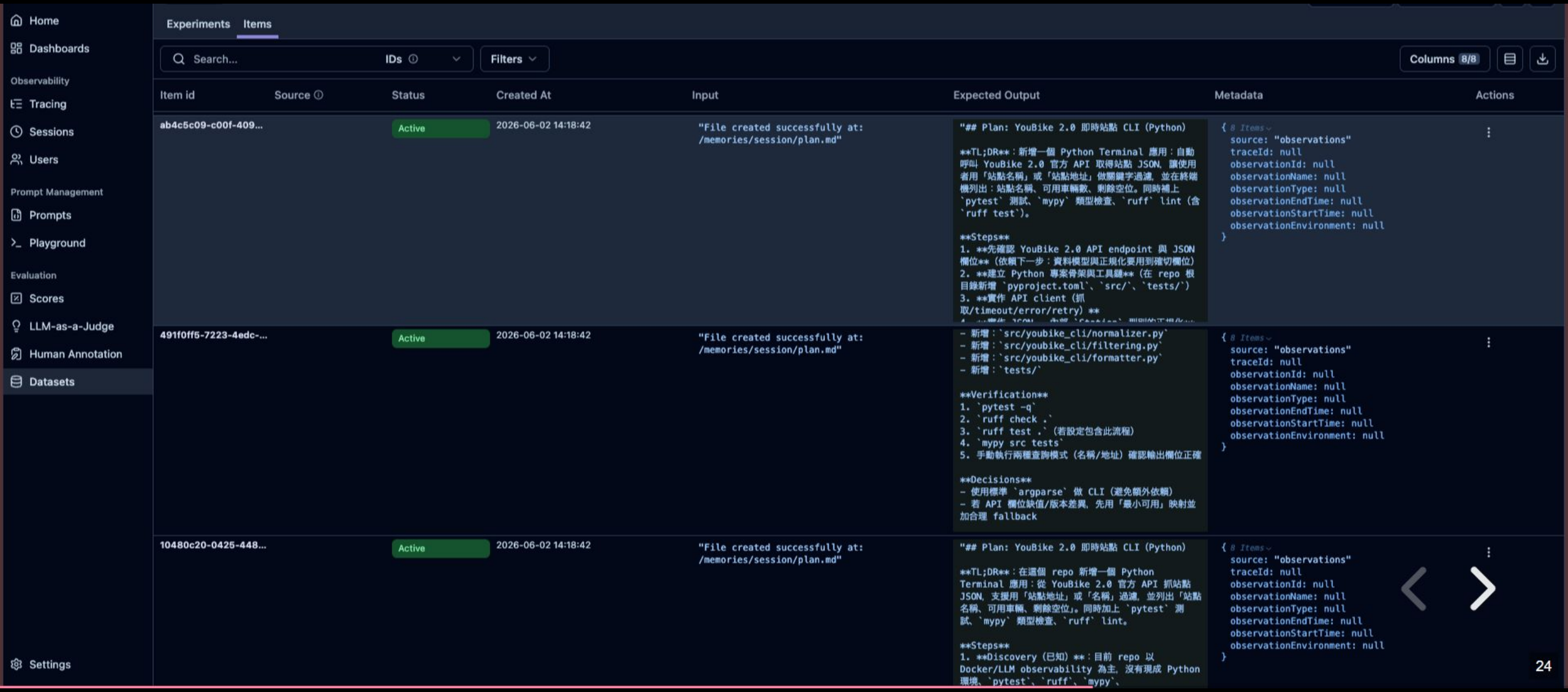
從 Daily Work Tracing 整理

```
Input：真實工作需求
Output：Model 的回答
Judge Score：回答品質
```

Experiment｜同一份考題

控制變因，重複執行

```
Model：5.4 → 5.5
Tools / Instruction：改動前後
```


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Experiments
Items
Search...
IDs
Filters
Columns 8/8
Item id
Source
Status
Created At
Input
Expected Output
Metadata
Actions
ab4c5c09-c00f-409...
Active
2026-06-02 14:18:42
"File created successfully at: /memories/session/plan.md"
"## Plan: YouBike 2.0 即時站點 CLI (Python)
TL;DR：新增一個 Python Terminal 應用：自動呼叫 YouBike 2.0 官方 API 取得站點 JSON，讓使用者用「站點名稱」或「站點地址」做關鍵字過濾，並在終端機列出：站點名稱、可用車輛數、剩餘空位。同時補上 `pytest` 測試、`mypy` 類型檢查、`ruff` lint（含 `ruff test`）。
Steps
1. **先確認 YouBike 2.0 API endpoint 與 JSON 欄位**（依賴下一步：資料模型與正規化要用到確切欄位）
2. **建立 Python 專案骨架與工具鏈**（在 repo 根目錄新增 `pyproject.toml`、`src/`、`tests/`）
3. **實作 API client（抓取/timeout/error/retry）**
{ 8 Items
source: "observations"
traceId: null
observationId: null
observationName: null
observationType: null
observationEndTime: null
observationStartTime: null
observationEnvironment: null
491f0ff5-7223-4edc-...
Active
2026-06-02 14:18:42
"File created successfully at: /memories/session/plan.md"
- 新增：`src/youbike_cli/normalizer.py`
- 新增：`src/youbike_cli/filtering.py`
- 新增：`src/youbike_cli/formatter.py`
- 新增：`tests/`
Verification
1. `pytest -q`
2. `ruff check .`
3. `ruff test .`（若設定包含此流程）
4. `mypy src tests`
5. 手動執行兩種查詢模式（名稱/地址）確認輸出欄位正確
Decisions
- 使用標準 `argparse` 做 CLI（避免額外依賴）
- 若 API 欄位缺值/版本差異，先用「最小可用」映射並加合理 fallback
{ 8 Items
source: "observations"
traceId: null
observationId: null
observationName: null
observationType: null
observationEndTime: null
observationStartTime: null
observationEnvironment: null
10480c20-0425-448...
Active
2026-06-02 14:18:42
"File created successfully at: /memories/session/plan.md"
"## Plan: YouBike 2.0 即時站點 CLI (Python)
TL;DR：在這個 repo 新增一個 Python Terminal 應用：從 YouBike 2.0 官方 API 抓站點 JSON，支援用「站點地址」或「名稱」過濾，並列出「站點名稱、可用車輛、剩餘空位」。同時加上 `pytest` 測試、`mypy` 類型檢查、`ruff` lint。
Steps
1. **Discovery（已知）**：目前 repo 以 Docker/LLM observability 為主，沒有現成 Python 環境、`pytest`、`ruff`、`mypy`、
{ 8 Items
source: "observations"
traceId: null
observationId: null
observationName: null
observationType: null
observationEndTime: null
observationStartTime: null
observationEnvironment: null
24

STEP 3｜整理 DATASET

- Plan：Context Precision
- Coding：Answer Correctness
- 依 Judge 分數分類
- 高品質案例作為改進參考

注意：從 O11y 走進Dataset，是個大坑

STEP 3 | DATA PREPROCESSING

- 清理 Tracing 格式，挑選所需 Features
- 例如 `Output.tool_calls.args`
- 記錄呼叫哪些 Tools、帶哪些參數
- 用 Langfuse SDK 轉換格式
- Partition Dataset

持久化儲存，作為測試題庫

STEP 3｜第三次優化：選擇最佳工具

- 用題庫測試不同選項：
  - Model：Pro、Mini、Nano
  - Model 版本：GPT-5.4、GPT-5.5
  - Thinking Effort
  - Agent：Codex、OpenCode、Copilot
- 不必等 Model 變強才解決問題
- 選適合的工具，平衡品質與成本

STEP 3 | EVALUATION → COST ESTIMATE

- 測試題數與使用的 Model
- LLM API Metrics、Token 用量
- Quality Estimate

沒有測試基礎的預算，只是市場喊價

STEP 0｜先釐清需求

需求應該先釐清

許多團隊卻先討論要不要換工具：

- 需求還不清楚
- 成員對 LLM 的理解有落差

STEP 0｜先釐清需求

需求應該先釐清

許多團隊卻先討論要不要換工具：

- 需求還不清楚
- 成員對 LLM 的理解有落差

先建立監測，讓團隊看見數據

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用數據調整認知，讓討論回到現實

STEP 0｜CODING AGENT 效率

想提升團隊效率，先定義「效率」：

$$\text{Overall Efficiency} = \frac{\text{產出數量} \times \text{產出品質}}{\text{Token Cost} \times \text{花費時間}}$$

降低 Cost、Latency、Error Rate
控制 Context Window
提升 Output Quality

STEP 0｜從需求走向 ACTION

依任務選工具：

- Model：
  - Mini、Nano
  - Low Latency、Low Error Rate
- Gateway Routing：
  - AgentGateway
  - LiteLLM
- Context Management：rtk
- Greenfield：Spec-kit

STEP 0｜讓團隊 ON THE SAME PAGE

- Dev Team：Chargeback
- 選擇 Model 與 Tool
- 與 Stakeholders、管理者對齊期待
- 量化效能指標：
  - 正確率、Error Rate
  - Latency、Cost
- 有 AI，不代表產出就能提升十倍

有 Evaluation 後，可以說：

> *產出約提升 2–3 倍，數據在此*

TAKEAWAY｜從觀察到決策系統

- Step 1：Baseline、第一次優化
- Step 2：LLM-as-a-Judge 評估
- Step 3：Dataset、Experiment
- Step 0：先釐清需求
- Decision：釐清需求 → Action

Q&A

- Lab | LLM O11y Decision System
- Lab | Spec-kit SDD